

Course Description and Outcome Form

Department of Computer Science and Engineering

School of Data and Sciences

Brac University

A. Course General Information:

Course Code:	CSE330
Course Title:	Numerical Methods
Credit Hours (Theory+Lab):	3+0
Contact Hours (Theory+Lab):	3+3
Category:	Program Core
Type:	Required
Prerequisites:	MAT216
Co-requisites:	None
Discord Server Link	https://discord.com/invite/xujGNttuy

B. Course Catalog Description (Content):

Fixed Point Arithmetic, Polynomial Interpolation, Differentiation, Nonlinear Equations, Linear Equations, Least-Squares Approximation, Numerical Integration

C. Course Objective:

The course will help students to recognize the need for numerical analysis, and the importance of error analysis. They will learn various methods to linearize a polynomial, differentiate and integrate different functions by using approximations, and finally how to solve the linearized equation by using the laws of linear algebra, like Gaussian elimination, QR decomposition, etc.

D. Faculty List:

Name	Initial	Section(s)
Abu Mohammad Khan	AMK	1, 2
Mushfique Nasir Probor	MNP	3, 4, 5, 6
Md Nafiu Rahman	NAFR	7, 8, 9
Md Fardin Rahman Ami	FDM	10
Aunon Halder	ANNH	11
Rifa Tasfiya	RFTS	12, 13, 14
Anindita Labonno	ALB	15, 16
Md. Aquib Azmain	AQU	17
Atonu Roy Chowdhury	NARC	18, 19

E. Course Outcomes (COs): Upon successful completion of this course, students will be able to

Sl.	CO Description	Weightage (%)
CO1	Demonstrate an understanding of the fundamental concept of numerical analysis for different mathematical problems and the need for error analysis.	10
CO2	Comprehend different numerical techniques and theorems of numerical methods for polynomial interpolation, derivatives, linear and non-linear equations, and integrations.	30
CO3	Apply the principles and/or techniques of numerical methods in different mathematical setups and solve the relevant mathematical problems.	60

F. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/ level	Delivery methods and activities	Assessment tools
CO1	Demonstrate an understanding of the fundamental concept of numerical analysis for different mathematical problems and the need for error analysis.	(b)	Cognitive/ Analyze	Lectures, Notes, Lab	Midterm, Final exams
CO2	Comprehend different numerical techniques and theorems of numerical methods for polynomial interpolation, derivatives, linear and non-linear equations, and integrations.	(c)	Cognitive/ Evaluate	Lectures, Notes, Lab	Midterm, Final exams
CO3	Apply the principles and/or techniques of numerical methods in different mathematical setups. Solve the linear and overdetermined systems by different numerical methods. Also compute integration numerically using Newton-Cotes formula and estimate the error.	(a)	Cognitive/ Analyze	Lectures, Notes, Lab	Midterm, Final exams

G. Course Materials:

i. Text and Reference Books:

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1.	Numerical Analysis II – Lecture Notes. (Main Text)	Anthony Yeates.	2018		Durham University	
2.	Numerical Methods in Engineering with Python.	Jaan Kiusalaas.	2005	First Edition	Cambridge University Press	ISBN-13: 978-0-521-8528 7-6

						ISBN-10:0-521-85287-0
3.	Numerical Analysis (Reference)	Richard L. Burden J. Douglas Faires	2011	Ninth Edition	BROOKS/C OLE CENGAGE Learning	ISBN-13:978-0-538-73351-9 ISBN-10:0-538-73351-9

ii. **Other materials (if any)**

(a) Lecture Notes/Handouts

H. Lesson Plan:

Lecture #	Topics
Lecture -1	Introduction and overview of the course. Fixed- and Floating-point numbers (Ch-1.1, 1.2) and their properties. Exponent bias. Significant figures. Examples.
Lecture -2	Rounding error (1.4), loss of Significance (1.5), examples.
Lecture -3	Interpolation definition and conditions, Single node or Taylor series interpolation (2.1), Taylor Theorem and its proof (using Roll's theorem).
Lecture -4	Multiple node interpolation: definition and conditions, Vandermonde method, Theorem-2.4 (2.2), examples. Quiz-1.
Lecture -5	Lagrange method: definition and properties of basis, interpolation polynomial (2.3), example. Assignment-1 due.
Lecture -6	Newton's method: definition and properties of basis and coefficients, Divided-Difference form (2.4)
Lecture -7	Proof of Divided-Difference formula, examples. Quiz-2.
Lecture -8	Interpolation error, Cauchy's theorem with proof (2.5), examples, Chebyshev nodes and relevance to Runge phenomena.
Lecture -9	Derivative conditions (2.7), Hermite basis and interpolation, example.
Lecture -10	Differentiation: forward, backward and central difference methods (definition and derivations), truncation error (Relative and Upper bound) (3.1). Assignment-2 due.
Lecture -11	Rounding error (3.2), Richardson extrapolation: properties and derivations (3.3), examples. Quiz-3.
Midterm Exam Week	The midterm exam for CSE330 will be held centrally on July 29, 2026 (Wednesday).

Lecture -12	Bisection method (4.1), examples, iteration number, multiple roots. Fixed-point function (definition and properties) (4.2).
Lecture -13	Construction of fixed-point functions, Ratio and convergence. Contraction Mapping Theorem. (4.2)
Lecture -14	Convergence rate λ , Types of convergences, examples, Newton's method, advantages and disadvantages of Newton's method.
Lecture -15	Quasi-Newton method: definition and usefulness, Secant method (4.4), Newton's method for systems (4.5). Quiz -4.
Lecture -16	Linear systems: definition and condition for unique solution, Triangular system and its properties, computational complexity (5.1). Assignment-3.
Lecture -17	Gaussian method (5.2) with examples, pivoting problem.
Lecture -18	<i>LU</i> -decomposition method with examples (5.3), Properties of Frobenius matrices.
Lecture -19	Orthonormal system and its properties. Least-square approximation methods (definition and properties), Discrete -square method (6.1, 6.2), example. Quiz -5.
Lecture -20	<i>QR</i> -decomposition method with example (6.3), Continuous least-square approximation method with example (6.4).
Lecture -21	Numerical Integration: concept, definition and properties, Newton-Cotes formula: Closed and Open (7.1), Trapezium rule, error calculation (Theorem-7.1).
Lecture -22	Simpson's rule, Composite Closed Newton-Cotes formula (7.2), example. Assignment-4.
Lecture -23	(if time permits): Exactness (7.3), Gaussian quadrature (7.4), and reviews. Quiz -6.
Final Exam	The final of CSE330 will be held centrally on September 16, 2026 (Wednesday).

I. Lab

No	Topic	Week/Lecture	Related CO
1	Floating Point and Basic Python	Week 1	C01
2	Polynomial [Class design] and application of Vandermonde Matrix	Week 2	C02
3.	Lagrange	Week 3	C02
4.	Hermite and Newton's Divided Difference	Week 4	C02
MID-WEEK			
5.	Differentiation and Richardson extrapolation	Week 5	C03

6.	Coding MID-Exam	Week 6	
7.	Root finding using Bisection method and Fixed-point iteration	Week 7	C03
8.	Equation Solving	Week 8	C03
9.	Integration	Week 9	C03
10.	Review Class	Week 10	
11.	Final Viva/Coding Exam	Week 11	

J. Assessment Tools:

Assessment Tools	Weightage (%)
Attendance	5
Quiz	15
Assignment	10
Midterm	25
The mandatory final exam	25
Lab	20

K. CO Assessment Plan:

Assessment Tools	Course Outcomes		
	CO1	CO2	CO3
Mid Term	√	√	√
Final	√	√	√

L. Makeup Exam Policy:

The university policy for makeup exams for the midterm and final exams are the following: “If a student misses a scheduled exam due to unforeseen circumstances, they must apply for a makeup exam within 10 days from the day of the exam. Within these 10 days, the student must collect the necessary approvals, including medical clearances, and submit the makeup form to the Office of the Controller of Examinations. **Students unable to appear in the final exams may sit for the final makeup exams on the dates decided by the department/school which will be within 4 weeks of the following semester.** Students failing to appear in the makeup final exams shall be awarded an ‘F’ Grade and shall have to retake the course”. For further details, visit the university webpage at <https://www.bracu.ac.bd/academics/office-registrar/policies-and-procedures/examinations>

M. Attendance Policy:

As per BRACU, attendance in class is **mandatory** for all undergraduate students. To be eligible for the **final assessment/examination**, undergraduate students must meet the following **minimum attendance requirements: (1) 90% attendance** for laboratory sessions for programs across the university, **(2) 70% attendance** at classes, other than laboratories, for programs across the university and **(3) 80% attendance** for studio sessions in the Department of Architecture. The Head of

Department, in consultation with the Registrar, will make arrangements to consider exceptions for students unable to attend classes because of major circumstances beyond their control.

N. Grading policy:

As per BRAC University Policy, the grades at the university will be indicated in the following manner:

97 - 100* = A+ (4.0) Exceptional

90 - <97* = A (4.0) Excellent

85 - <90 = A- (3.7)

80 - <85 = B+ (3.3)

75 - <80 = B (3.0) Good

70 - <75 = B- (2.7)

65 - <70 = C+ (2.3)

60 - <65 = C (2.0) Fair

57 - <60 = C- (1.7)

55 - <57 = D+ (1.3)

52 - <55 = D (1.0) Poor

50 - <52 = D- (0.7)

<50 = F (0.0) Failure

O. Course Coordinators:

Theory Sections : Abu Mohammad Khan (AMK) and Md. Aquib Azmain (AQU).

Lab Sections : Anindita Labonno (ALB).